

PRACTICAL 1: NETWORK TOOLS AND SAFETY

Crimping Tool- Attaches RJ45 connectors to network cables

Safe handling.

Keep fingers away from the crimping mechanism and store safely after use

Cable Tester- Checks whether a network cable is wired correctly and functioning

Safe handling.

Test only low-voltage network cables and handle gently to avoid damage

Punch Down Tool- Inserts and secures wires into patch panels or wall sockets

Safe handling.

Use the correct size and apply controlled pressure

Wire Stripper- Removes insulation from network cables without damaging internal wires

Safe handling.

Cut away from the body and keep blades covered when not in use

RJ45 Connector- Connects network cables to devices such as computers, switches, and routers

Safe handling.

Handle carefully and avoid bending or damaging the pins

Network Cable (UTP)- Transfers data between network devices

Safe handling.

Avoid sharp bends, pulling, or placing heavy objects on the cable

Screwdriver Set -Opens or secures networking equipment

Safe handling.

Use the correct screwdriver size and keep hands dry

Multimeter- Measures electrical properties for basic troubleshooting

Safe handling.

Follow operating instructions and avoid testing live circuits unless trained

Health and Safety Procedures

1. Wear appropriate personal protective equipment (PPE) where required.
2. Keep the work area clean and free of clutter.
3. Switch off equipment before installing or removing cables.
4. Avoid exposed wires and damaged equipment.

5. Do not eat or drink near networking equipment.
6. Report faulty tools and equipment immediately.
7. Always follow laboratory or workplace safety regulations.

Tool Name	Purpose/Use	Condition	Quantity
Crimping Tool	Attaches RJ45 connectors to network cables	Good	1
Cable Tester	Tests continuity and wiring of cables	Good	1
Punch Down Tool	Connects wires to patch panels and keystones	Good	1
Wire Stripper	Removes cable insulation	Good	1
Network Screwdriver Set	Tightens and loosens screws on network devices	Good	1 Set
RJ45 Connectors	Connect network cables to devices	Good	20
Ethernet Cables	Provide network connectivity	Good	10

5. Health and Safety Procedures

When working with networking tools and equipment:

1. Wear appropriate personal protective equipment (PPE) where required.
2. Keep the work area clean and free from clutter to prevent trips and falls.
3. Inspect tools before use and do not use damaged tools.
4. Switch off and unplug equipment before installing or servicing hardware.
5. Handle cables carefully to avoid damaging them or causing hazards.
6. Use tools only for their intended purpose.
7. Follow electrical safety procedures when working near powered equipment.
8. Store tools safely after use.
9. Report any faulty equipment or safety hazards immediately.
10. Always follow workplace health and safety policies.